

File: mL_algo_from_scratch/README.md

Purpose: Linear Regression implementation for continuous value prediction

=== Execution Block (Main) ===

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# mL_algo_from_scratch
```

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Simple implementations of common ML algorithms from scratch, organized as classes (object-oriented).

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Contents:

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- `knn.py` - K-Nearest Neighbors (classification/regression)
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Math Alert: This involves calculus! Derivatives are used to update weights.

```
- `linear_regression.py` - Linear regression (closed-form and gradient descent)
```

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- `gradient_descent.py` - small gradient descent helper
```

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- `naive_bayes.py` - Gaussian Naive Bayes classifier
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- `utils.py` - helper functions
```

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- `examples/run_examples.py` - demonstrates usage on synthetic data
```

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Requirements:

=== Execution Block (Main) ===

- numpy

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Run examples:

=== Execution Block (Main) ===

```
```bash
```

### === Execution Block (Main) ===

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```
python -m mL_algo_from_scratch.examples.run_examples
```

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